

CSS Corner Shape

Complete Guide: Easy English Edition

Master rounded corners and custom border shapes in modern CSS

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1 Introduction

CSS corner shape properties let you round, curve, or customize the corners of elements. From simple rounded corners to complex shapes, you have complete control over how your elements look!

1.1 What is Corner Shape?

Corner shape refers to how the corners of an element appear:

- **Sharp Corners:** Default square corners (0px radius)
- **Rounded Corners:** Curved corners (border-radius)
- **Custom Shapes:** Advanced corner effects

1.2 Why Customize Corners?

- **Modern Design:** Rounded corners feel softer and friendlier
- **Visual Hierarchy:** Different shapes draw attention
- **Brand Identity:** Unique corner styles match your brand
- **User Experience:** Rounded elements feel more approachable
- **Accessibility:** Softer designs can be easier on the eyes

2 Border-Radius Basics

2.1 What is Border-Radius?

`border-radius` controls how curved the corners of an element are.

2.1.1 Basic Syntax

```
border-radius: [radius];

/* Examples */
border-radius: 8px;           /* All corners */
border-radius: 50%;           /* Perfect circle */
border-radius: 10px 20px;     /* Horizontal, Vertical */
```

2.1.2 Common Values

Value	Effect	Example
0	Sharp corners	<code>border-radius: 0;</code>
4px	Slight curve	<code>border-radius: 4px;</code>
8px	Medium curve	<code>border-radius: 8px;</code>
16px	More rounded	<code>border-radius: 16px;</code>
50%	Perfect circle	<code>border-radius: 50%;</code>

2.2 All Four Corners

Round all four corners equally:

```
.box {
  border-radius: 8px;
  background: white;
  padding: 20px;
}
```

2.3 Individual Corners

Control each corner separately:

```
/* Syntax: top-left, top-right, bottom-right, bottom-left */
border-radius: 8px 16px 8px 16px;

/* Or use specific properties */
border-top-left-radius: 8px;
```

```
border-top-right-radius: 16px;
border-bottom-right-radius: 8px;
border-bottom-left-radius: 16px;
```

2.3.1 Shorthand Patterns

Pattern	Meaning
<code>border-radius: 8px;</code>	All four corners = 8px
<code>border-radius: 8px 16px;</code>	TL/BR = 8px, TR/BL = 16px
<code>border-radius: 8px 16px 4px;</code>	TL=8px, TR/BL=16px, BR=4px
<code>border-radius: 8px 16px 4px 12px;</code>	TL=8px, TR=16px, BR=4px, BL=12px

2.4 Practical Examples

Example 1: Card Component

```
.card {
  border-radius: 8px;
  background: white;
  box-shadow: 0 2px 8px rgba(0,0,0,0.1);
  padding: 20px;
}
```

Soft, modern card design.

Example 2: Button

```
.button {
  border-radius: 4px;
  background: #2196F3;
  color: white;
  padding: 10px 20px;
  border: none;
  cursor: pointer;
}
```

Example 3: Avatar Circle

```
.avatar {
  border-radius: 50%;
  width: 64px;
  height: 64px;
  object-fit: cover;
}
```

Perfect circle for user avatars.

3 Advanced Border-Radius

3.1 Elliptical Corners

Create non-circular rounded corners:

```
/* Elliptical: horizontal / vertical */  
border-radius: 50% 50% 0 0; /* Rounded top, square bottom */  
border-radius: 10px 20px / 5px 15px; /* Different curves */
```

3.1.1 Example: Pill Shape (Horizontal Ellipse)

```
.pill {  
  border-radius: 25px;  
  padding: 10px 20px;  
  background: #2196F3;  
  color: white;  
  display: inline-block;  
}
```

3.1.2 Example: Dome Shape (Vertical Ellipse)

```
.dome {  
  border-radius: 50% / 100% 100% 0 0;  
  width: 200px;  
  height: 100px;  
  background: #2196F3;  
}
```

3.2 Creating Organic Shapes

```
/* Organic rounded shape */  
.blob {  
  border-radius: 60% 40% 50% 60% / 40% 60% 50% 40%;  
  width: 200px;  
  height: 200px;  
  background: #2196F3;  
}
```


4 Corner Shape Properties

4.1 border-top-left-radius

Round only the top-left corner:

```
.element {  
  border-top-left-radius: 16px;  
}
```

4.2 border-top-right-radius

Round only the top-right corner:

```
.element {  
  border-top-right-radius: 16px;  
}
```

4.3 border-bottom-right-radius

Round only the bottom-right corner:

```
.element {  
  border-bottom-right-radius: 16px;  
}
```

4.4 border-bottom-left-radius

Round only the bottom-left corner:

```
.element {  
  border-bottom-left-radius: 16px;  
}
```

5 Real-World Examples

5.1 Example 1: Modern Card Design

```
.card {  
  border-radius: 12px;  
  background: white;  
  overflow: hidden;  
  box-shadow: 0 4px 12px rgba(0,0,0,0.1);  
  transition: transform 0.3s ease;  
}  
  
.card:hover {  
  transform: translateY(-4px);  
}  
  
.card-image {  
  width: 100%;  
  height: 200px;  
  object-fit: cover;  
}  
  
.card-content {  
  padding: 20px;  
}
```

5.2 Example 2: Rounded Button Group

```
.button-group {  
  display: flex;  
  border-radius: 8px;  
  overflow: hidden;  
}  
  
.button {  
  flex: 1;  
  padding: 10px 20px;  
  border: none;  
  background: #E0E0E0;  
  cursor: pointer;  
  transition: background 0.3s;  
}  
  
.button:first-child {  
  border-radius: 8px 0 0 8px;  
}
```

```
.button:last-child {
  border-radius: 0 8px 8px 0;
}

.button:hover {
  background: #2196F3;
  color: white;
}
```

5.3 Example 3: Modal with Rounded Corners

```
.modal {
  border-radius: 12px;
  background: white;
  box-shadow: 0 20px 60px rgba(0,0,0,0.3);
  padding: 30px;
  max-width: 500px;
}

.modal-title {
  font-size: 24px;
  margin-bottom: 15px;
}

.modal-close {
  position: absolute;
  top: 10px;
  right: 10px;
  border-radius: 50%;
  width: 32px;
  height: 32px;
  border: none;
  cursor: pointer;
}
```

5.4 Example 4: Notification Toast

```
.toast {
  border-radius: 8px;
  padding: 16px 20px;
  background: white;
  box-shadow: 0 4px 12px rgba(0,0,0,0.15);
  margin-bottom: 10px;
  display: flex;
  align-items: center;
  gap: 12px;
}

.toast.success {
  border-left: 4px solid #4CAF50;
}

.toast.error {
  border-left: 4px solid #F44336;
}
```

```
}  
  
.toast.warning {  
  border-left: 4px solid #FF9800;  
}
```

6 Tips and Tricks

6.1 Smooth Corners with Overflow Hidden

```
.box {  
  border-radius: 8px;  
  overflow: hidden; /* Hides content that spills outside corners */  
  background: white;  
}
```

6.2 Animating Border-Radius

```
.interactive {  
  border-radius: 8px;  
  transition: border-radius 0.3s ease;  
}  
  
.interactive:hover {  
  border-radius: 50%; /* Animate to circle */  
}
```

6.3 Creating Different Shapes

6.3.1 Circle

```
.circle {  
  border-radius: 50%;  
  width: 100px;  
  height: 100px;  
}
```

6.3.2 Pill/Capsule

```
.pill {  
  border-radius: 25px;  
  padding: 10px 20px;  
}
```

6.3.3 Half-Circle

```
.half-circle {  
  border-radius: 100px 100px 0 0;  
  width: 200px;  
}
```

```
    height: 100px;  
}
```

6.3.4 Teardrop

```
.teardrop {  
    border-radius: 50% 50% 50% 0;  
    width: 100px;  
    height: 100px;  
}
```

6.4 Using Different Units

```
/* Pixels */  
border-radius: 8px;  
  
/* Percentage */  
border-radius: 50%;  
  
/* Rem (relative to root font size) */  
border-radius: 0.5rem;  
  
/* Em (relative to element font size) */  
border-radius: 0.5em;
```

7 Browser Support

7.1 Compatibility

Browser	Support
Chrome	All versions
Firefox	All versions
Safari	All versions
Edge	All versions
Opera	All versions
IE 9	Partial support (with prefix)
IE 8	NOT SUPPORTED

7.2 Vendor Prefixes (Legacy)

For older browsers, use prefixes:

```
.element {  
  -webkit-border-radius: 8px; /* Safari, Chrome (old) */  
  -moz-border-radius: 8px; /* Firefox (old) */  
  border-radius: 8px; /* Standard */  
}
```

8 Common Mistakes

8.1 Mistake 1: Border-Radius on Images

DON'T do this:

```
/* Won't work as expected with images */
img {
  border-radius: 8px;
}
```

DO this instead:

```
/* Wrap image in a container */
.image-wrapper {
  border-radius: 8px;
  overflow: hidden;
  display: inline-block;
}

.image-wrapper img {
  display: block;
  width: 100%;
}
```

8.2 Mistake 2: Forgetting Overflow Hidden

Problem:

```
.box {
  border-radius: 8px;
  background: white;
  box-shadow: 0 0 10px rgba(0,0,0,0.3);
}

.box-content {
  background: #FF0000;
  /* Content sticks out! */
}
```

Solution:

```
.box {
  border-radius: 8px;
  overflow: hidden;
  box-shadow: 0 0 10px rgba(0,0,0,0.3);
}
```


8.3 Mistake 3: Too Many Values

DON'T do this:

```
/* Maximum 4 values! */  
border-radius: 8px 12px 16px 10px 14px; /* INVALID */
```

DO this:

```
/* Maximum 4 values, or use individual properties */  
border-radius: 8px 12px 16px 10px;  
border-top-left-radius: 8px;
```

9 Advanced Combinations

9.1 Mixing with Box-Shadow

```
.card {  
  border-radius: 12px;  
  background: white;  
  box-shadow:  
    0 2px 4px rgba(0,0,0,0.1),  
    0 4px 8px rgba(0,0,0,0.1),  
    0 8px 16px rgba(0,0,0,0.1);  
}
```

9.2 With CSS Gradients

```
.gradient-card {  
  border-radius: 12px;  
  background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);  
  padding: 20px;  
  color: white;  
}
```

9.3 With Transforms

```
.interactive-card {  
  border-radius: 12px;  
  transition: all 0.3s ease;  
}  
  
.interactive-card:hover {  
  border-radius: 20px;  
  transform: scale(1.05) rotate(1deg);  
}
```

9.4 With Borders

```
.bordered-card {  
  border-radius: 12px;  
  border: 2px solid #2196F3;  
  padding: 20px;  
  background: white;  
}
```

10 Responsive Corner Radius

10.1 Adjust Corners for Screen Size

```
.card {  
  border-radius: 8px;  
  background: white;  
  padding: 20px;  
}  
  
@media (min-width: 768px) {  
  .card {  
    border-radius: 12px;  
  }  
}  
  
@media (min-width: 1024px) {  
  .card {  
    border-radius: 16px;  
  }  
}
```

10.2 Fluid Border-Radius

```
.responsive-card {  
  border-radius: clamp(8px, 5vw, 24px);  
  padding: 20px;  
  background: white;  
}  
  
/* Scales smoothly between 8px and 24px */
```

11 Quick Reference

11.1 Common Border-Radius Values

Value	Use Case	Example
0	Sharp corners	Desktop apps
2px-4px	Subtle curve	Business apps
8px	Default modern	Most designs
12px-16px	Rounded look	Modern web
50%	Circle	Avatars, buttons

11.2 Border-Radius Syntax Summary

```
/* All four corners */
border-radius: 8px;

/* Top-left, bottom-right / top-right, bottom-left */
border-radius: 8px 16px;

/* Top-left / top-right, bottom-right / bottom-left */
border-radius: 8px 16px 12px;

/* Top-left / top-right / bottom-right / bottom-left */
border-radius: 8px 16px 12px 10px;

/* Individual corners */
border-top-left-radius: 8px;
border-top-right-radius: 16px;
border-bottom-right-radius: 12px;
border-bottom-left-radius: 10px;

/* Elliptical (horizontal / vertical) */
border-radius: 50% 50% 0 0;
```

11.3 Key Properties

- **border-radius** - All corners
- **border-top-left-radius** - Top-left only
- **border-top-right-radius** - Top-right only
- **border-bottom-right-radius** - Bottom-right only
- **border-bottom-left-radius** - Bottom-left only

11.4 Units

- **px** - Pixels (fixed size)
- **%** - Percentage (relative to element size)
- **em/rem** - Relative to font size
- **vw/vh** - Viewport width/height

Conclusion

CSS corner shape properties, especially `border-radius`, are essential for modern web design. They're simple to use but powerful enough to create beautiful, sophisticated designs.

Key Takeaways:

- Use `border-radius: 8px;` as a modern default
- Use `border-radius: 50%;` for circles and rounded buttons
- Remember to use `overflow: hidden;` for smooth corners
- Individual properties for fine-tuned control
- Combine with animations for interactive effects
- Supported in all modern browsers

Start using rounded corners in your designs today. They instantly make websites look more modern and friendly!